

QUESTION PAPER - SET 4

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If

9. yes, share details.

10. Tell us about your family.

11. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

In the late 21st century, the Global Council enacted the "Resource Equilibrium Accord" to address escalating conflicts over dwindling freshwater reserves. The Accord introduced a tiered quota system, assigning water access rights based on each nation's ecological footprint, technological capability for desalination, and per-capita consumption history.

While wealthier nations invested in advanced desalination plants and atmospheric water harvesters, poorer nations — despite having smaller ecological footprints — struggled to meet their citizens' basic needs under their reduced quotas.

Critics argued that the Accord unintentionally entrenched inequality, as historical consumption patterns favored nations that had previously overused water.

To address the backlash, an amendment proposed the creation of a "Water Equity Fund" financed by a surcharge on nations exceeding their quota. The fund would support sustainable water projects in under-resourced regions.

Yet, debates continued over enforcement: how to monitor usage accurately, prevent data manipulation, and ensure that the fund genuinely benefited those in need.

1. What was the main goal behind the introduction of the Resource Equilibrium Accord in the late 21st century?

- A. To establish a global water trading market
- B. To balance global water distribution and prevent conflicts over freshwater resources
- C. To reward nations with low per-capita water consumption
- D. To increase funding for desalination research

2. Which aspect most contributed to the Resource Equilibrium Accord unintentionally reinforcing inequality among nations?

- A. Technological advancements in desalination
- B. Inaccurate monitoring of water usage
- C. Historical consumption patterns that benefited countries with prior high water use
- D. Opposition to the Water Equity Fund

3. Which conclusion is best supported by the information provided in the passage about the Resource Equilibrium Accord?

- A. The amendment eliminated all concerns about fairness
- B. Nations with smaller ecological footprints were given larger quotas
- C. Wealthier nations opposed atmospheric water harvesting
- D. The Accord unintentionally reinforced pre-existing inequalities

4. Under the Resource Equilibrium Accord, when would a nation be required to pay a surcharge to support the Water Equity Fund?

- A. When its per-capita consumption is below the global average

- B. When it fails to reduce its ecological footprint
- C. When it does not participate in atmospheric water harvesting
- D. When its water usage exceeds the assigned quota

5. What is the main challenge that remains unresolved regarding the amendment to the Resource Equilibrium Accord?

- A. Lack of sustainable water project ideas
- B. The unwillingness of wealthier nations to pay surcharges
- C. Uncertainty in the fund's revenue source
- D. Ensuring transparency and preventing misuse of funds

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. Choose the correct transformation for the sentence: "The weather was too rough for the pilot to board."

- A. The weather was so rough that the pilot could board.
- B. The weather was rough enough for the pilot to board.
- C. The weather was so rough that the pilot could not board.
- D. The weather was very rough but the pilot boarded.

2. Choose the correct passive voice for the sentence: "The crew completed the cargo operation successfully."

- A. The cargo operation was completed successfully by the crew.
- B. The cargo operation had completed successfully by the crew.
- C. The cargo operation is completed by the crew successfully.
- D. The cargo operation was being completed successfully.

3. Identify the grammatically correct sentence from the options below.

- A. The captain emphasized the importance of punctuality.
- B. The captain emphasized on the importance of punctuality.
- C. The captain emphasized about the importance of punctuality.
- D. The captain emphasized for the importance of punctuality.

4. She can solve navigation problems as quickly as _____ officer on board.

- A. Many
- B. Every
- C. Any
- D. Some

SECTION 5 - NUMERICAL APTITUDE

1. Two workers can complete a job in 10 hours and 15 hours respectively. If they work together, how long will it take them to finish the job?

- A. 6 hours
- B. 9 hours
- C. 12 hours
- D. 7 hours

2. If the total number of crew is 120, how many crew members belong to the Engine department?

- A. 60
- B. 45
- C. 40
- D. 50

3. Choose the correct option in the sequence.

- A. Option A
- B. Option B
- C. Option C
- D. Option D

4. A person needs to walk from one corner of a square park to the opposite corner. If he walks along the edges, he covers the length of two sides, but if he walks diagonally, he takes the shortest route.

Approximately what percentage of distance does he save by walking diagonally instead of along the edges?

- A. 29%
- B. 24%
- C. 33%
- D. 20%

5. Two boats are anchored on opposite sides of a 100 m tall lighthouse. From the boats, the angles of elevation to the top of the lighthouse are 30° and 45° respectively. What is the distance between the two boats?

- A. 273 m
- B. 173 m
- C. 200 m
- D. 300 m

SECTION 6 - MARITIME KNOWLEDGE

1. What does the term “lee shore” refer to in seamanship?

- A. A shore protected from wind
- B. A shore facing the wind
- C. A dangerous shore onto which wind blows
- D. A shallow coastal area

2. On a nautical chart, which color is commonly used to indicate shallow water and the boundaries of certain regulated areas?

- A. Gold
- B. Blue
- C. Black
- D. Green

3. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?

- A. Cold front
- B. Occluded front
- C. Warm front
- D. Stationary front

4. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

5. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

- A. Anticlockwise and inward
- B. Clockwise and inward
- C. Clockwise and outward
- D. Anticlockwise and outward

6. Which cloud type is most commonly associated with thunderstorms and heavy rainfall?

- A. Cirrus
- B. Stratus
- C. Cumulonimbus
- D. Altocumulus

7. According to ship safety regulations, which of the following must be made weathertight as part of the Conditions of Assignment?

- A. Hatchways situated on the freeboard deck
- B. Sounding pipes on deck
- C. Cargo hold ventilators below deck
- D. Scuppers above deck

8. The azimuth of a celestial body is measured at which of the following locations?

- A. Observer's Zenith
- B. Observer's Nadir
- C. Observer's Meridian
- D. Observer's Prime Vertical

9. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

10. The total area beneath the GZ (righting arm) curve from 0° to 40° of heel is a measure of which of the following vessel characteristics?

- A. The vessel's dynamic stability within that heel range
- B. The vessel's metacentric height (GM)
- C. The vessel's KM
- D. The vessel's maximum righting lever

11. What does the term "angle of loll" describe in ship stability?

- A. The angle at which the center of gravity is vertically above the center of buoyancy
- B. The time taken for a vessel to complete one roll cycle
- C. KM is greater than GM
- D. The heel angle at which a vessel with negative GM achieves equilibrium due to hull shape

12. What is the purpose of a ship's whistle or horn?

- A. Entertainment
- B. Distress communication only
- C. Navigational sound signaling
- D. Engine warning

13. Which gas is primarily responsible for the greenhouse effect?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Helium

14. What is the function of a ship's rudder?

- A. To increase stability
- B. To change the ship's direction
- C. To reduce rolling
- D. To measure speed

15. Which type of knot is commonly used to secure a rope to a post?

- A. Reef Knot
- B. Bowline
- C. Clove Hitch
- D. Sheet Bend

16. What is the primary purpose of a ship's radar?

- A. Cooking assistance
- B. Detecting objects and navigation
- C. Measuring cargo weight
- D. Monitoring fuel usage

17. At which location on Earth does the Coriolis effect reach its greatest strength?

- A. At the Equator
- B. At the Tropics
- C. At 30° latitude
- D. At the Poles

18. What is the main purpose of a ship's anchor?

- A. To increase vessel speed
- B. To hold the vessel in position
- C. To improve fuel efficiency
- D. To reduce rolling motion

19. Which part of the ship is located at the front?

- A. Stern
- B. Bow
- C. Port
- D. Starboard

20. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker
- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

21. The observer's prime vertical passes through the east and west points of which of the following?

- A. Rational horizon
- B. Equinoctial
- C. Celestial meridian
- D. Great circle

22. What does "starboard" refer to on a ship?

- A. Left side
- B. Front side
- C. Right side
- D. Rear side

23. What is the primary purpose of a marine sextant?

- A. Measuring wind direction
- B. Determining celestial altitude
- C. Measuring sea depth
- D. Determining engine power

24. What does "port side" mean on a vessel?

- A. Right side
- B. Left side
- C. Front side
- D. Upper deck

25. A vessel alters course from 090° to 180°. What is the amount of alteration?

- A. 45°
- B. 90°
- C. 180°
- D. 270°

26. Which instrument is used to determine the depth of water beneath a vessel?

- A. Barometer
- B. Echo Sounder
- C. Sextant
- D. Hygrometer

27. Ventilators forming part of Conditions of Assignment must:

- A. Be of any height if fitted with wire mesh
- B. Have minimum height and efficient closing appliances
- C. Have closing devices only above superstructure decks
- D. Be permanently open for machinery ventilation

28. Which navigation light is carried by a vessel not under command?

- A. Two all-round red lights in a vertical line
- B. One all-round white light
- C. Green and red sidelights only
- D. Two all-round green lights

29. What is the primary purpose of a ship's freeboard?

- A. To increase cargo capacity
- B. To improve engine efficiency
- C. To provide reserve buoyancy and safety
- D. To reduce fuel consumption

30. When crossing a traffic separation scheme, the vessel should:

- A. Cross at any convenient angle
- B. Cross at right angles to the traffic flow
- C. Alter course to avoid lanes
- D. Follow the traffic flow